

GuideCursor AI

Browser AI Copilot — Project Documentation

1. Project Overview

GuideCursor AI is a browser extension that attaches a secondary 'mini cursor' beside the user's mouse. It visually guides users through web interfaces without automating their actions — acting as a **teacher cursor**, not a bot.

How it works

- User presses **Ctrl + Shift + A** on any webpage
- A popup offers Text or Voice input
- User asks: *"I can't find the search button"*
- Extension captures screenshot + DOM + element coordinates
- AI analyzes the page and returns structured JSON steps
- Mini cursor detaches, animates to the target, highlights it, then returns

2. Problem & Objectives

Problems with current web navigation

- UI designs change frequently
- Beginners struggle with navigation
- Existing AI assistants automate actions or open chat windows
- Interrupts workflow and limits accessibility

Key objectives

- Attach and animate a mini cursor alongside the user's cursor
- Accept voice and text commands via Web Speech API
- Capture page screenshots + DOM structure for AI analysis
- Generate and render step-by-step visual guidance
- Response time target: <2 seconds
- Lightweight, cross-browser, secure

3. System Architecture

Layer	Component	Responsibilities
Browser Extension	Cursor, Listener, DOM Extractor	Mini cursor, keyboard shortcut, screenshot, DOM capture
Backend API	Node.js / Express	Auth, request handling, AI routing
AI Service	GPT-4 Vision / Gemini	Image analysis, UI understanding, step generation

Frontend Renderer	React + Framer Motion	Cursor animation, highlights, step messages
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4. Technology Stack

Category	Technologies
Frontend	React, TypeScript, Tailwind CSS, Framer Motion
Browser APIs	Extension API, DOM APIs, Web Speech API
Backend	Node.js, Express.js (optional: NestJS)
Database	MongoDB or PostgreSQL
Speech-to-Text	Whisper API
Vision / AI	GPT-4 Vision, Gemini Vision

5. AI JSON Response Format

The AI returns a structured JSON object with page intent and coordinate-based steps:

```
{
  "page": "youtube.com",
  "intent": "find_search",
  "steps": [
    { "step": 1, "instruction": "Move to top area" },
    { "step": 2, "instruction": "Look at center navigation bar" },
    { "step": 3, "instruction": "Search bar located here",
      "coordinates": { "x": 850, "y": 72 }, "highlight": true }
  ]
}
```

6. Key Challenges & Solutions

Browser security restrictions

→ Use content scripts to safely interact with pages

Large screenshot size

→ Compress images before sending to reduce latency

Dynamic website UIs

→ Combine screenshot + DOM analysis for robustness

AI hallucination / wrong coordinates

→ Validate AI coordinates against actual DOM elements

7. MVP Build Phases

Phase	Deliverable
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Phase 1	Browser extension + mini cursor following original cursor
Phase 2	Keyboard shortcut (Ctrl+Shift+A) + popup UI
Phase 3	Voice input via Web Speech API
Phase 4	Screenshot + DOM capture
Phase 5	AI backend integration + analysis
Phase 6	JSON guidance output
Phase 7	Animated cursor guidance + highlights

8. Design Note

Avoid relying on screenshots alone. Combine DOM structure + screenshot + element coordinates. This makes AI guidance significantly more accurate since screenshots miss interactive context.

Future Features

- Multi-language voice commands
- Accessibility mode
- Gesture control
- Personalized learning & AI memory
- Tutorial mode
- Eye tracking support
- Mobile browser support